

The Landau-Selberg-Delange method for Dirichlet L -functions, and applications

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General problem: Given a Dirichlet series $\sum_n a_n/n^s$ with natural analytic properties, give suitable estimates on $\sum_{n \leq x} a_n$.

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$$\sum_{n \leq x} a_n = \frac{x}{(\log x)^{1-\alpha}} \sum_{j=0}^N \frac{\kappa_j}{(\log x)^j} + O(\text{Error Terms}) \quad (1)$$

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Extension? Assume $\sum_n a_n/n^s$ “behaves like” $\prod_{\chi \bmod q} L(s, \chi)^{\alpha_\chi}$, for some $\{\alpha_\chi\}_\chi \subset \mathbb{C}$.

- Give as **precise asymptotic series** estimating $\sum_{n \leq x} a_n$, **uniformly in q in a wide range** (wider than from previous literature).

One of the main results

Theorem 1 (S.R. 2025, outline).

Assume $\sum_n a_n/n^s$ “behaves like” $\prod_{\chi \bmod q} L(s\nu, \chi)^{\alpha_\chi}$, for some $\{\alpha_\chi\}_\chi \subset \mathbb{C}$ and some fixed $\nu \in \mathbb{R}^+$. Then for any fixed $K_0 > 0$,

$$\sum_{n \leq x} a_n = \frac{x^{1/\nu}}{(\log x)^{1-\alpha_{\chi_0}}} \sum_{0 \leq j \leq N} \frac{\kappa_j}{(\log x)^j} + O(\text{Error Terms}),$$

uniformly in $x \geq 3$, $N \geq 0$ and $q \leq (\log x)^{K_0}$.

Error term: Genuine saving over main term in the full range $q \leq (\log x)^{K_0}$ in several applications of interest.

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Wider ranges under Landau-Siegel zeros conjecture and GRH.

Some applications

- (1) Positive integers with prime divisors restricted to residue classes:
Given $q \in \mathbb{Z}^+$ and $\mathcal{A} \subset (\mathbb{Z}/q\mathbb{Z})^\times$, estimate
 $\#\{n \leq x : p \mid n \implies p \bmod q \in \mathcal{A}\}.$

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- (2) Sathe–Selberg Theorem in arithmetic progressions:
- Sathe–Selberg: Local distributions of the functions
 $\omega(n) = \sum_{p \mid n} 1$ and $\Omega(n) = \sum_{p \mid n} \nu_p(n)$.

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- Generalization:

$$\omega_a(n) := \sum_{\substack{p \mid n \\ p \equiv a \bmod q}} 1 \quad \text{and} \quad \Omega_a(n) := \sum_{\substack{p \mid n \\ p \equiv a \bmod q}} \nu_p(n).$$

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- Theorem 1 $\implies$ Local distributions of $\omega_a(n)$ and $\Omega_a(n)$ modulo $q \leq (\log x)^{K_0}$.

One more application for now

- (3) **Residue-class distribution of multiplicative functions:** Given multiplicative functions $f_1, \dots, f_K : \mathbb{Z}^+ \rightarrow \mathbb{Z}$, estimate $\#\{n \leq x : (\forall i) f_i(n) \equiv a_i \pmod{q}\}$ for units $a_i \pmod{q}$.

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✓ Large classes of $f_1, \dots, f_K$ and **fixed** q .
 - **Theorem 1** $\implies$ ✓ Extend to $q \leq (\log x)^{K_0}$
✓ Qualitatively + Quantitatively **optimal** results.

Thank you for your attention!

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